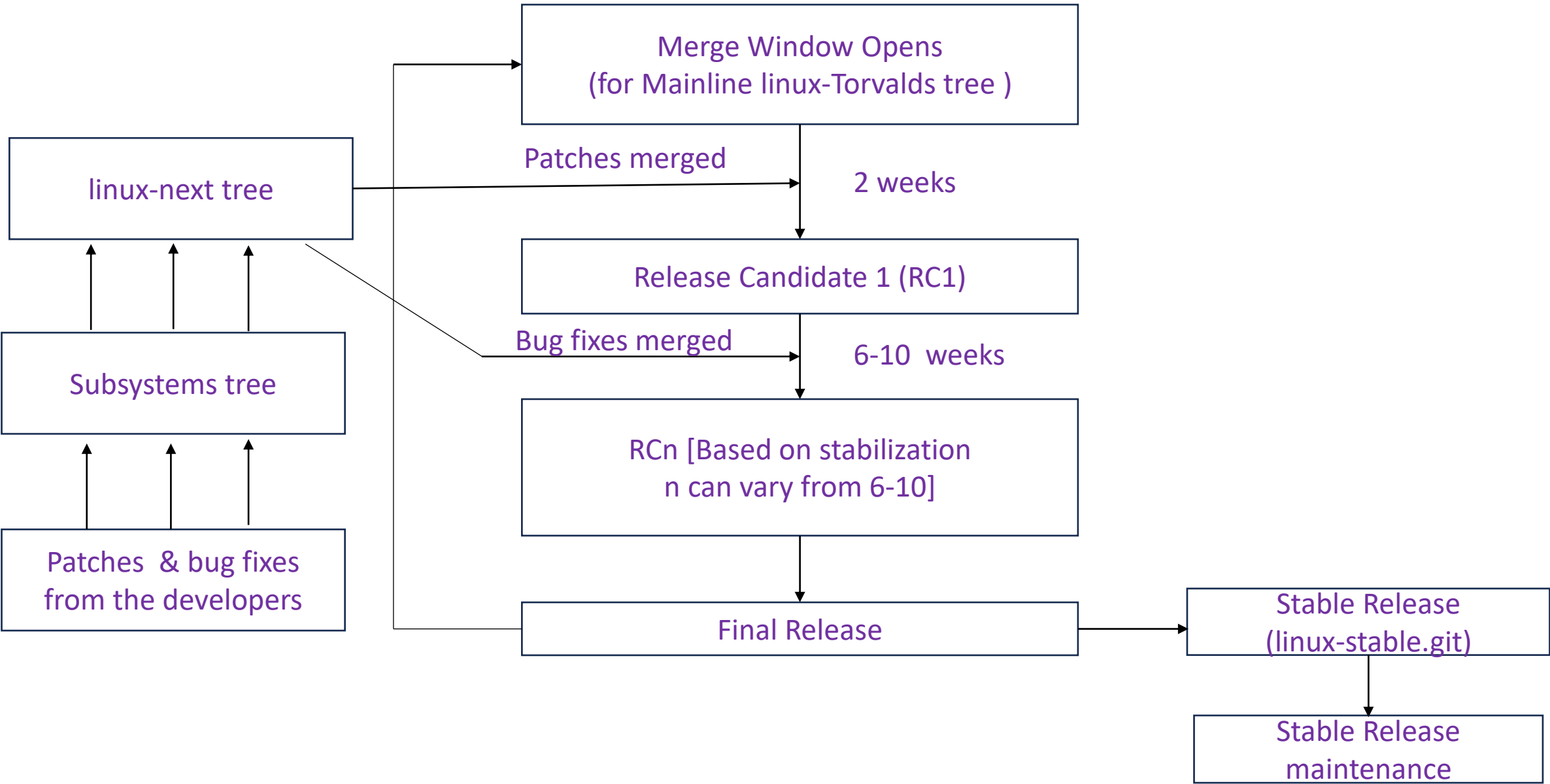


Linux Kernel Development Process

Infobell IT Solutions

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Linux Upstream (Mainline) Development Cycle



Linux patch upstream development process

Upstream patch development steps:

- Describe the problem being solved and plans of implementation.
- Work is being done in a corporate environment requires managers permission to be posted in public mailing list.
- Send the plans and the implementation details to the list of Linux kernel subsystem maintainers and request for comments.
- After getting official buy-in from maintainers, start the implementation.
- Implementation:
 - ☐ Follow the coding guidelines (coding style, inline functions, locking, etc)
 - ☐ Give attention to compile time warnings/errors
 - ☐ Use code checking tools to find static & runtime issues.
 - ☐ Check for regressions.
 - ☐ Add appropriate documentation.
- Preparing the patch for posting
 - ☐ If the changes are large, break into logical sequences and each contains most simple changes.
 - ☐ Then prepare each sequence as patch in logical order. Now it becomes series of patches for posting.
 - ☐ A one-line description of what the patch does. A blank line followed by a detailed description of the contents of the patch.
 - ☐ One or more tag lines, with, at a minimum, one Signed-off-by: line from the author of the patch.
- Post the patch for review to the relevant subsystem maintainers and CC to open mailing list and developers who are already working on similar issues.
- Address the review comments given by the reviewers.
- Once the patch(es) are approved, they will be merged into the subsystem maintainers tree.
- Inclusion of patches into subsystem trees will get attention from other developers may trigger further review comments. The review comments should be addressed in same way.
- From the subsystems tree, the patches will go to linux-next tree. From there, they will go to mainline tree, when the merge window opens.